ELK Stack

01. <u>ELK Stack Architecture</u>

Overview

The ELK Stack (Elasticsearch, Logstash, Kibana, and Beats) is designed to collect, process, store, search, and visualize large volumes of machine-generated data, particularly logs.

It solves key challenges faced in Security Operations Centers (SOC):

- High log volume and velocity
- Diverse log formats
- Fast search requirements
- Detection and investigation needs

ELK enables near real-time log analytics using a distributed and scalable design.

Core Architectural Philosophy

ELK follows a pipeline-driven architecture:

Data Producers → Data Shippers → Processing Layer → Storage & Search Engine → Visualization Layer

Example flow:

Firewall Logs → Filebeat → Logstash → Elasticsearch → Kibana

Each component performs a dedicated function, ensuring modularity and scalability.

Core Components & Responsibilities

Elasticsearch — Storage & Query Engine

Elasticsearch acts as the central data store and search engine.

Key responsibilities:

- Stores structured data as JSON documents
- Indexes events for fast retrieval
- Performs full-text search and aggregations
- Supports distributed scaling

Important concepts:

- Index → Logical data container
- Document → Individual event/log entry
- Shard → Horizontal partition of data
- Replica → Redundant copy for availability

Elasticsearch is optimized for search-heavy workloads rather than transactional operations.

Logstash — Data Processing Engine

Logstash functions as the transformation and enrichment layer.

Pipeline model:

Input → Filter → Output

Key responsibilities:

- Parses incoming logs
- Normalizes fields
- Enriches data (GeoIP, metadata, etc.)
- Routes events to destinations

Logstash enables consistent data structures required for correlation and detection.

Kibana — Visualization & Investigation Layer

Kibana provides the user interface for interacting with Elasticsearch data.

Key capabilities:

- Dashboards & visualizations
- Log searching & filtering
- Time-series analysis
- Alerting & monitoring

In SOC workflows, Kibana supports threat hunting and anomaly detection.

Beats — Lightweight Data Shippers

Beats are lightweight agents used for data collection.

Examples:

- Filebeat → Log files
- Winlogbeat → Windows Event Logs
- Metricbeat → System metrics
- Packetbeat → Network traffic

Advantages:

- Low resource usage
- Reliable log delivery
- Backpressure handling

Data Flow Model

ELK data flow typically follows:

1. Logs generated by sources
2. Beats collect & forward events
3. Logstash processes & transforms data
4. Elasticsearch indexes events
5. Kibana visualizes & analyzes data

This separation of concerns improves performance and flexibility.

Scaling & Distribution Model

Elasticsearch distributes data across nodes using:

- Shards → Partitioning
- Replicas → Fault tolerance

Benefits:

- High availability
- Horizontal scalability
- Improved query performance

Clustered deployments prevent single points of failure.

02. ELK Stack Network Configurations & Protocols

Why Networking Matters

ELK Stack is a distributed system where components communicate continuously.

Proper networking ensures:

- Reliable log ingestion
- Stable cluster communication
- Secure data transfer
- Performance optimization

Incorrect network configurations can break pipelines or degrade performance.

Component Communication Paths

Beats → Logstash

- Protocol → TCP
- Default Port → 5044

Beats use TCP for reliable delivery and flow control.

Example:

Filebeat → TCP 5044 → Logstash

Logstash → Elasticsearch

- Protocol → HTTP / HTTPS
- Default Port → 9200

Used for:

- Indexing events
- Query operations
- Cluster interactions

HTTPS is preferred in secure environments.

Kibana → Elasticsearch

- Protocol → HTTP / HTTPS
- Default Port → 9200

Kibana acts as a client querying Elasticsearch APIs.

Common ELK Ports

Component	Port	Purpose
Elasticsearch	9200	REST API / Queries
Kibana	5601	Web Interface
Logstash Input	5044	Beats Ingestion

Knowledge of ports is critical for troubleshooting and firewall rules.

Protocols Used in Log Pipelines

ELK environments commonly use:

- TCP → Reliable transport
- UDP → Syslog ingestion
- HTTP / HTTPS → API-based ingestion
- Syslog → Network device logs

Example:

Firewall → Syslog (UDP 514) → Logstash

Security Considerations

Production-grade deployments secure communications using:

- TLS encryption
- Authentication & authorization
- Firewall restrictions
- Node-to-node encryption

Unsecured pipelines expose sensitive log data.

03. ELK Stack Integrations

Importance of Integrations

ELK Stack's effectiveness depends entirely on data ingestion.

Without meaningful log sources, ELK cannot support detection or analysis workflows.

SOC visibility requires multi-source integrations.

Common Log Sources

Typical integrations include:

- Firewalls
- IDS / IPS systems
- Endpoint agents
- Application servers
- Cloud platforms

Each source generates logs in unique formats.

Integration Mechanisms

ELK supports multiple ingestion methods:

- Beats agents
- Syslog forwarding
- REST APIs
- Custom scripts / pipelines

Example:

Winlogbeat → Windows Logs → Elasticsearch

Log Normalization Challenges

Log formats vary significantly across systems:

- Field names differ
- Timestamp formats vary
- Message structures differ

Logstash solves this using:

- Grok filters
- Field mutations
- Data enrichment

Normalization is essential for correlation and detection accuracy.

SOC-Relevant Considerations

- Integration quality impacts detection capability
- Incorrect parsing leads to false insights
- Consistent field structures enable analytics

04. ELK Stack Deployment Models

Single Node Deployment

All components run on a single machine.

Suitable for:

- Learning environments
- Testing & labs
- Small workloads

Limitations:

- No fault tolerance
- Limited scalability
- Performance constraints

Multi Node / Cluster Deployment

Elasticsearch nodes distributed across infrastructure.

Benefits:

- High availability

- Fault tolerance
- Horizontal scaling
- Improved performance

Used in production SOC environments.

Elasticsearch Node Roles

Common node types:

- Master Node → Cluster coordination
- Data Node → Stores indexed data
- Ingest Node → Pre-processing pipelines

Separating roles improves cluster efficiency.

Cloud vs On-Prem Deployment

ELK can be deployed:

- On-premises (self-managed)
- Cloud-hosted (managed services)

Trade-offs include cost, control, and maintenance complexity.

05. Alternatives to ELK Stack

Why Alternatives Exist

While ELK Stack is flexible and cost-effective, it requires tuning and management.

Alternatives provide:

- Built-in detections
- Vendor support
- Simplified administration

Splunk

Strengths:

- Powerful analytics engine
- Mature ecosystem
- Enterprise adoption

Limitations:

- High licensing costs

Graylog

- Built on Elasticsearch
- Simplified log management
- User-friendly interface

Wazuh

- Open-source SIEM platform
- Often integrated with ELK
- Strong endpoint security focus

ELK vs Commercial SIEM

ELK:

- Flexible
- Cost-efficient
- Highly customizable

Commercial SIEM:

- Prebuilt detections
- Easier management
- Vendor-supported

Alternatives to other

1️⃣ Alternatives to Kibana (Visualization & Analysis Layer)

🔷 Grafana : Used for:

✓ Dashboards ✓ Metrics visualization ✓ Multi-source data support ✓ Works with Elasticsearch

Limitation: ❌ Not a full ELK-native experience like Kibana

🔷 Graylog UI

Graylog provides its own interface.

✓ Search & dashboards ✓ Alerting ✓ Built on Elasticsearch

🔷 OpenSearch Dashboards

If using OpenSearch (Elasticsearch fork):

✓ Kibana-like interface ✓ AWS-backed ecosystem

2️⃣ Alternatives to Logstash :

🔷 Fluentd : Extremely popular in modern architectures.

✔ Lightweight & fast ✔ Strong Kubernetes adoption ✔ Flexible log routing

🔷 Fluent Bit Optimized version of Fluentd.

✔ Very low memory footprint ✔ Edge / container environments

3️⃣ Alternatives to Elasticsearch (Storage & Search Engine)

🔷 OpenSearch :

✔ Fork of Elasticsearch ✔ API-compatible ✔ Strong AWS adoption

🔷 Splunk :

✔ Full SIEM + analytics platform ✔ Very powerful search ✔ Expensive

🔷 ClickHouse :

✔ Extremely fast analytics ✔ Used for large-scale telemetry